making capabilities to operate in unstructured human environments will make the robots smart."

Cobots for industrial applications

A collaborative robot, or cobot, works interactively with humans in a shared workspace. It helps the human operator(s) to fulfill tasks and minimise risk in situations such as transportation and handling of sharp, pointed, hot or heavy work pieces.

Cobots are easy to use, flexible and safe. These are easy to program and can learn independently using machine learning. Instructions can also be given using a graphic user interface other than coding. Anyone can flexibly program cobots for a variety of tasks.

Wilson adds, "Application areas of smart robots are mostly in human environments. These days, cobots are increasingly used in production environments, because these can augment the capabilities of humans to work safely hand-in-hand with them.

"Another big application area is consumer robotics, which includes robots that can work in human homes."

Cobots versus industrial robots. Cobots are designed to work along with human operators, while industrial ones work in place of humans. Cobots are capable of self-learning during a job, while the latter require an engineer to write new code for any change in process.

Cobots are not designed for heavy manufacturing as these work closely with humans. Hence, these are safe enough to function around humans.

Aeolus smart home robot cleaning the floor (Credit: www.cnet.com)

Industrial robots, on the other hand, can handle heavier and larger materials, and require fencing to keep humans out of the workspace. Cobots immobilise at the slightest touch due to sophisticated sensors and, thus, prevent any danger to nearby people.

Developing smart robots

Building smart robots requires automation, AI and machine learning, for human-robot collaboration. Integrated sensors in robots have made it possible to handle assembly tasks accurately.

A smart robot detects its environment, learns from it and responds accordingly. To detect the environment, it requires sensors like lidar, temperature, depth, proximity and camera. The sensors interact with the environment in real time and generate the required information and responses. The robot checks the information using various algorithms to generate the required responses as per the situation or scenario. It then decides how to act.

Robot testing a smartphone (Credit: www.sastrarobotics.com)

Pradeep Shoran, assistant general manager - marketing, Kuka Robotics, says, "We need to understand the requirements of a customer and then build our solution around it. The trend is moving towards greater customisation, more product variants, away from rigid mass production. Hence, personalised products that a customer can self-configure on a computer and order on the Internet are in demand.

"Production of goods need to become considerably more flexible in this new world. To be able to achieve this, new production concepts are required that enable extremely versatile production on an industrial scale and networked throughout the process chain. Production logistics will undergo fundamental changes. Robots will move freely around the production facility on mobile platforms, performing a wide range of different tasks and interacting with humans."

Applications of smart robots

The smart robot market is growing rapidly with the adoption of autonomous AI-enabled robots for personal and professional services. Applications that utilise smart robots can perform the tasks better, faster and accurately. Various fields of application of smart robots include aerial (drones), industrial (heavy-duty automated machines) and underwater robotics, robotics for intelligent transportation systems, scientific research, manufacturing, mining, agriculture, construction, search-and-rescue operations, space research, medical assistance and personal assistance.

Shoran says, "Human-robot collaboration and mobile robotics are prominent emerging trends in robotics. For example, LBR iiwa is an autonomous, mobile, safe, sensitive and lightweight robot, controllable over a mobile platform. It is based on human-robot collaboration and works as a colleague for its operator, directly supporting the latter. It can independently fetch tools from the warehouse and load those into the tool magazines on the machines without additional safety equipment. The operator has unhindered access to machines. If the production sequence changes, the route taken by the autonomous vehicle changes without altering the